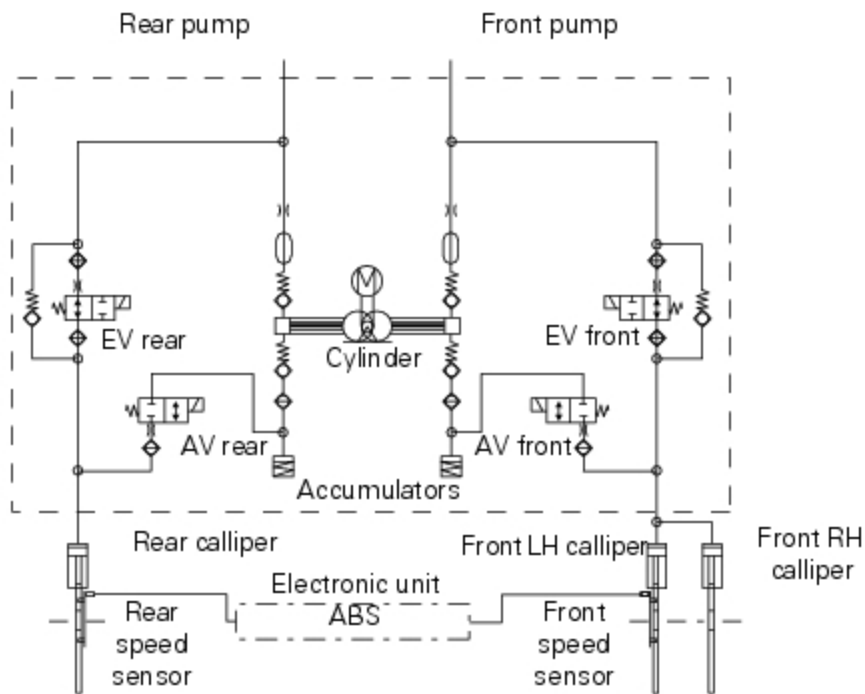


9.1 -ABS system information

Operating principle

The Ducati ABS brake system manages the front and rear brakes separately. A pulse generator (phonic wheel), with a ring of slots, is fixed onto each wheel. On the left calliper mounting bracket of the front fork and on the rear brake calliper holder plate are HALL effect sensors which detect the full sections and voids on the phonic wheel as the vehicle moves, providing an instantaneous wheel speed reading. These sensors transmit the data to the ABS control unit, which contains software implementing a specific control algorithm developed by Ducati. The software compares the average speed of the vehicle against the instantaneous wheel speed values to determine the degree of skidding. If the pressure applied to the calliper by the rider causes the control limits to be exceeded, the control unit intervenes in the braking circuit relative to the wheel for which incipient lock-up is detected. The ABS modulates the calliper pressure with a system of solenoid valves that initially prevents any further increase in hydraulic pressure (EV valve closed), and then reduces pressure (AV valve opened). The AV valve is opened in a series of pulses (with less than 10 milliseconds between successive pulses), to reduce pressure in steps. When the wheel begins to rotate again in response to the diminished braking force applied and the wheel speed reaches the reference value, the AV release valve is closed. Simultaneously, the EV inlet valve is reopened, restoring normal operation of the brake system. The ABS control unit can monitor and modulate brake force accordingly in the three following different conditions: dry road surface (high grip), wet or greasy road surface (poor grip) and uneven road surface. ABS functionality is disabled at vehicle speeds lower than 5 km/h.

A diagram illustrating how the ABS system functions is given below.



The hydraulic component of the ABS system consists of a primary circuit (from the cylinder to the control unit and from the control unit to the calliper) and a secondary circuit (completely within the control unit). The hydraulic layout of the ABS system is given below.

Key for ABS hydraulic system layout

EV rear	Rear calliper inlet valve
AV rear	Rear calliper outlet valve
EV front	Front calliper inlet valve
AV front	Front calliper outlet valve

ABS system operating information

The response of the system is based on the analysis of the speed signals for front and rear wheels; the system is automatically deactivated if either of these signals is missing.



Note

In the event of the ABS control unit detecting a fault in the ABS electronic management system, it activates the specific fault warning indicator on the instrument panel and restores conventional braking functionality without ABS. Hydraulic faults in the brake system and faults not directly related to the ABS system (e.g.: worn brake pads) cannot be detected by the ABS control unit.



Important

The response of the system is based on the values read for the front and rear wheel speeds; take great care not to damage the phonic wheels and relative speed sensors when removing the wheels or when working in the vicinity of these components; damage to the phonic wheels damages may compromise the operation of the system and cause dangerous malfunctions.

ABS diagnosis

The diagnostic function of the ABS 8M ascertains the functionality of the main system components via hardware/software tests, but cannot modify the operating parameters of the ABS system. The DDS diagnostic instrument is connected by cable to the ABS control unit communication link connector (A), located beneath the saddle. The connection cable between the tester and the ABS control unit has another branch with a negative and a positive clamp that must be connected to the vehicle battery to power the tester itself and enable communication with the ABS control unit.

From the AUTODIAGNOSIS menu of the DDS instrument, select the ABS system model and wait until the correct configuration loads for analysing the configuration of the vehicle. After switching the vehicle to key-on state, the tester communicates with the ABS control unit.

Once in this configuration, a number of parameters and states relative to the ABS control unit and information concerning the control unit itself may be selected and displayed.

If the tester receives notification from the ABS control unit that there are diagnostic errors dating from a prior test session still stored in its memory, the user is warned by specific message on the display and an audible alarm.

The ABS control unit identification number, identified by the prefix ID ECU, may be viewed from the INFO menu.

Select the function “View parameters and states”

This function allows the tester to display the following ABS system parameters and states:

Viewable parameters

Front wheel speed	The speed value for the front wheel is displayed in Km/h.
Rear wheel speed	The speed value for the rear wheel is displayed in Km/h.
System voltage value	The system voltage value, measured across the terminals of the ABS unit, is displayed in Volts

Viewable states

Hydraulic pump motor state	Indicates if the electric motor of the hydraulic pump inside the ABS unit is functioning or not.
Valve relay state	Displays the state of the main valve relay inside the ABS unit.
Rear input valve state	Displays the state of the normally open EV1 rear inlet valve (OFF if open, ON if closed).
Front input valve state	Displays the state of the normally open EV4 rear inlet valve (OFF if open, ON if closed).
Front output valve state	Displays the state of the normally closed AV4 rear outlet valve (OFF if closed, ON if open).
ABS switch state	Displays state of the analogue input of the ABS unit for the strategy switch off request (1 state to Uz, 0 state to GND).

Select the function “View errors”

This tester function provided the user with information concerning the error list in the ABS control unit memory, indicating if errors have been stored previously or if they are current. The following codes are given after a short description of the diagnosis type: MEM, indicating that the error has been stored previously by the ABS control unit but has not been detected in the current test session. ATT, indicating that the error is current and has been detected during the current test session.



Note

While ATT indicates that the error has been diagnosed during the current test session, it does not necessarily indicate that the error is actually active at the time of indication. For example: disconnecting the front wheel speed sensor causes the code ATT to be displayed after the error description, but the code continues to be displayed even once the cause of its activation has been eliminated by reconnecting the sensor, as the ABS will only check sensor operation again after the next key-off/key-on cycle. As a result, always perform a key-on/key-off cycle after any work on the ABS system, interrupting and reopening communication between the diagnostic instrument and the ABS control unit before checking the updated ABS error list again.

The abbreviations for all the errors displayed are given as follows:

Errors	Description
Front ABS speed sensor signal	Disconnected/short circuit to GND/short circuit to Uz of rear wheel speed sensor
Plausibility of signal from front wheel speed sensor	Front wheel speed sensor fault – Plausibility
Rear speed sensor signal	Disconnected/short circuit to GND/short circuit to Uz of front wheel speed sensor
Plausibility of signal from rear wheel speed sensor	Rear wheel speed sensor fault – Plausibility
Plausibility of difference between front and rear wheel speeds	Speed difference between front and rear wheel (WSS_GENERIC)
Front inlet valve fault	Front inlet valve (EV) fault
Rear outlet valve fault	Rear outlet valve (AV) fault
Rear inlet valve fault	Rear inlet valve (EV) fault
Front outlet valve fault	Front outlet valve (AV) fault
ABS pump fault	ABS pump motor fault
Valve relay fault	Valve relay fault (Failsafe relay)
ABS unit electronic control unit fault	ECU fault
ABS unit power circuit fault	ABS power circuit fault

Select the function “ABS system information”

This tester function displays the identification data for the ABS system, such as Supplier reference, Software version and BARCODE.

e.g.:

Supplier ref.: DUCM696

Software version: 67890

Bar Code Data: HQOJM98200RA

ABS system deactivation

Carry out the procedure described in section Sect. P 7, [ABS disabling function](#).



Warning

If the vehicle front wheel remains off the ground for a prolonged period while the vehicle is in motion, the speed difference detected between the two wheels causes an ABS fault (warning light activated) and makes it impossible for the control unit to establish a reference speed.